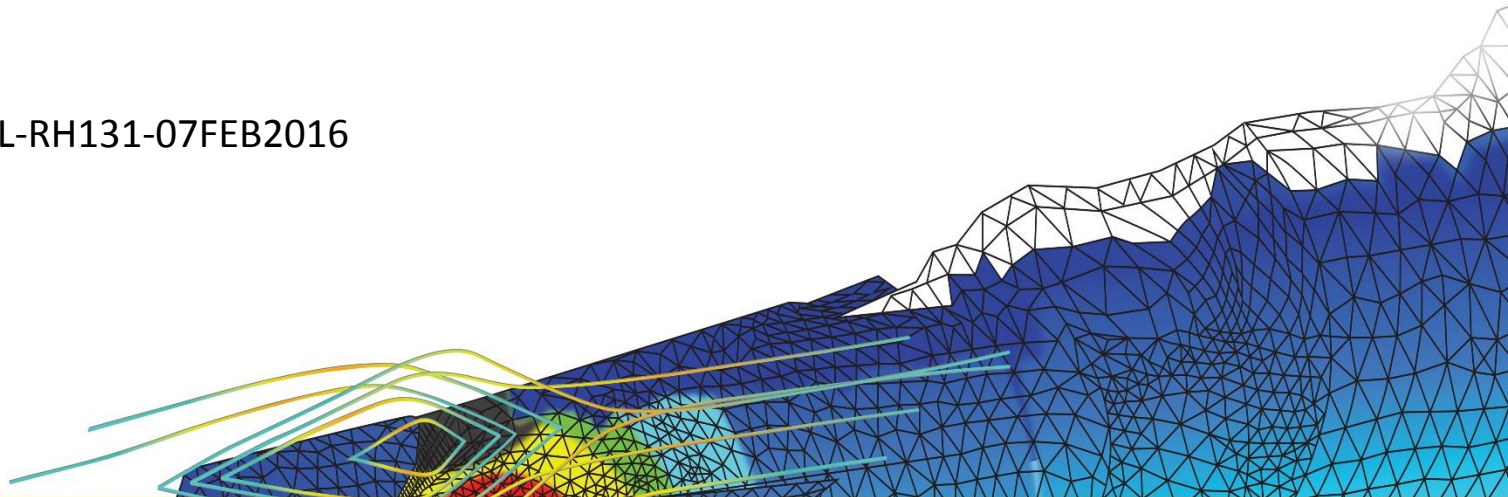




RedHawk™ VCD Based Dynamic Analysis

Version V5.2-AL-RH131-07FEB2016

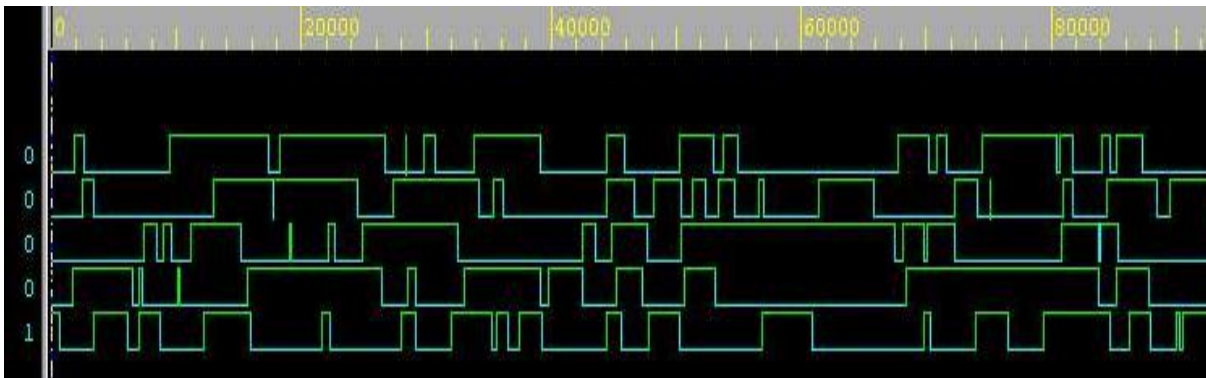


Agenda

- **VCD Introduction**
- **RedHawk VCD Based Static Flow**
- **Cycle Selection Basics**
- **RedHawk VCD Based Dynamic Flow**

What is VCD ?

- **VCD or “Value change dump file”**
 - Contains the waveforms for different signal nets in the design
 - Created from 3rd party verification tools
 - Format is ASCII
 - Waveform can be viewed in 3rd party tools
- **Alternative format is FSDB (Fast signal Database)**
 - Format is binary
 - Much lesser file size compared to VCD
- **RedHawk supports both VCD and FSDB formats**



```

$version
Tool:          APACHE          05.10.001-s
$end

$timescale
    1ps
$end

$scope module test_bench1 $end

$scope module GENERIC $end

$var wire      1 ||| net_63041 $end
$var wire      1 ||" net_18967 $end
$var wire      1 ||# net_15348 $end
$var wire      1 ||$ net_18964 $end
$var wire      1 ||% net_18963 $end
$var wire      1 ||& net_18962 $end
$var wire      1 ||' net_63034 $end
$var wire      1 ||( net_99214 $end
$var wire      1 ||) net_18958 $end
$var wire      1 ||* net_18957 $end
$var wire      1 ||+ net_18956 $end
$var wire      1 ||, net_18955 $end
$var wire      1 ||- net_18954 $end
$var wire      1 ||. net_18953 $end
$var wire      1 ||\ net_16165 $end
$var wire      1 ||\ net_16163 $end
$var wire      1 ||^ net_16161 $end
$var wire      1 ||_ net_16159 $end
$var wire      1 ||~ net_16157 $end
$var wire      1 ||a net_16155 $end
$var wire      1 ||b net_16153 $end
$var wire      1 ||c net_16151 $end
$var wire      1 ||d net_16149 $end
$var wire      1 ||e net_16147 $end
$var wire      1 ||f net_16145 $end
$var wire      1 ||g net_16143 $end
$var wire      1 ||h net_16054 $end
$upscope $end

$upscope $end

$enddefinitions $end

$dumpvars
0|||
1||"
1||#
1||$
0||%
0||&
1||'
1||(
1||)
1||*
1||+
0||,
0||-
1||.

```

Types of VCD

Gate Level VCD

- **Contains toggle information of all the gates in the design**
- **True time VCD**
 - SDF (Standard Delay Format) is used in verification
 - VCD generated from timing annotated functional/scan simulation
 - RedHawk will use switching scenario from VCD
 - RedHawk will use instance switching time from VCD
- **Non-true time VCD**
 - SDF not used in verification
 - VCD generated assuming unit-delay gate level functional/scan simulation
 - RedHawk will use switching scenario from VCD
 - RedHawk will use switching time from STA file
- **Do I need STA file for true-time based analysis ?**
 - Yes, you will still require this for slew information

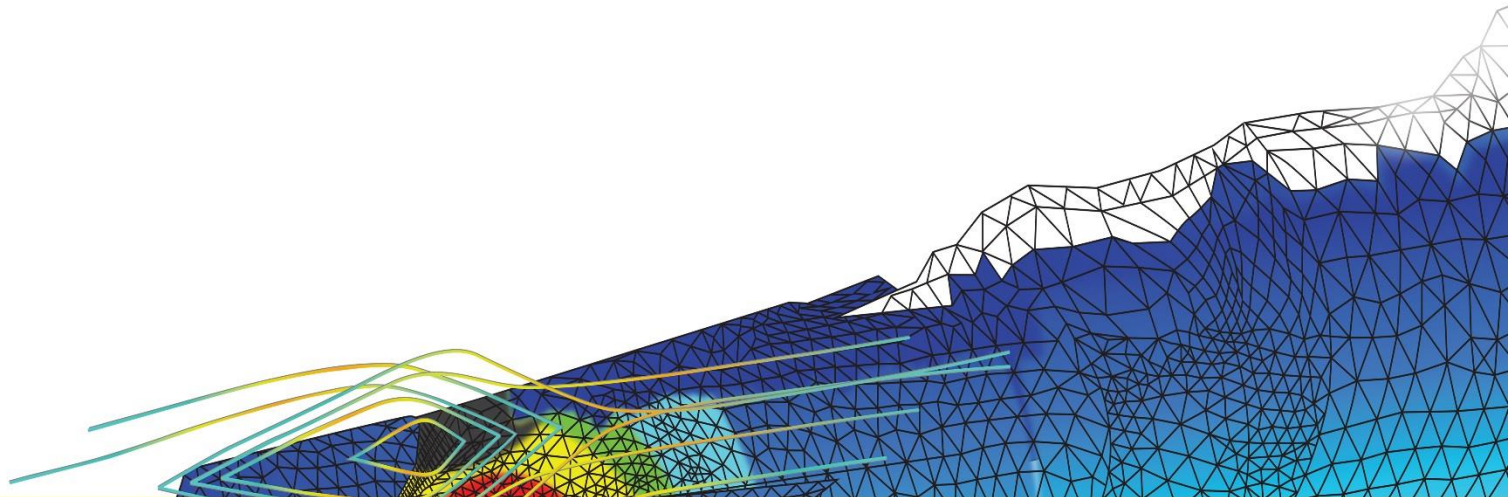
Types of VCD (*Cont'd*)

RTL VCD

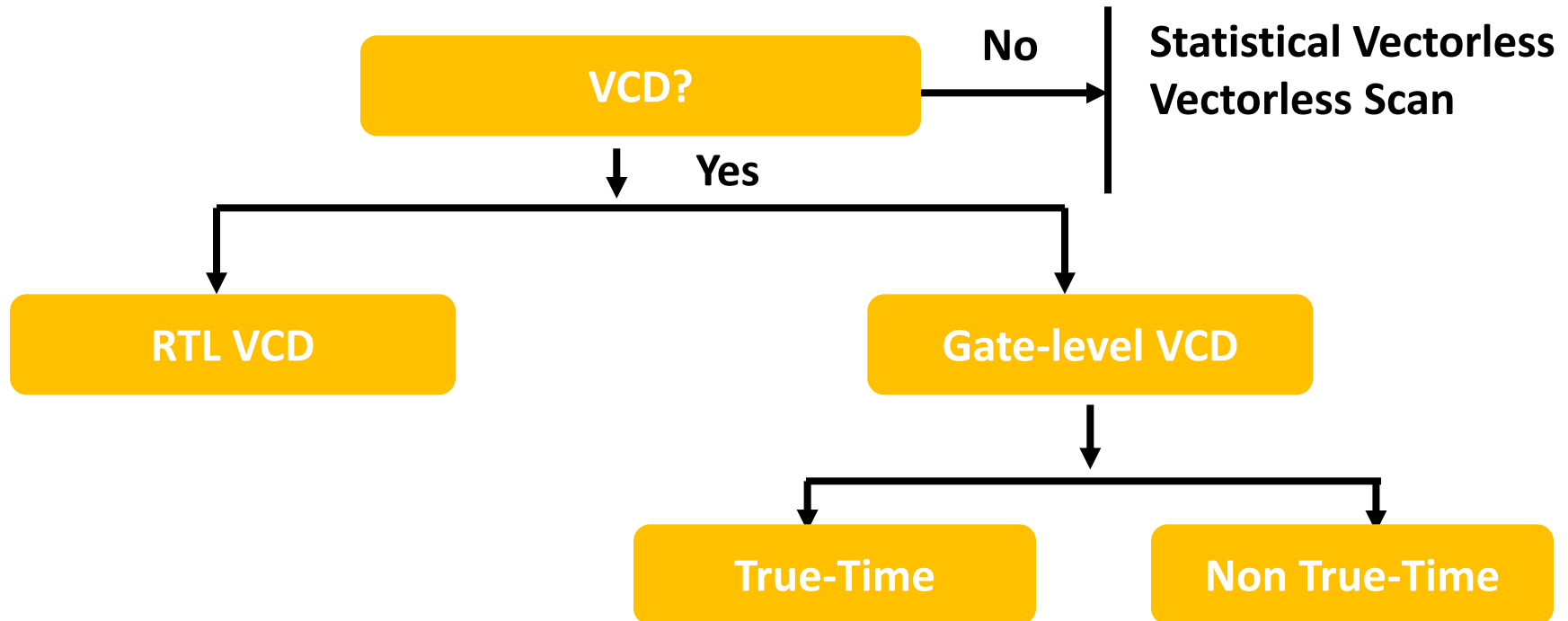
- Contains only activity at state points
 - Registers/latches/memory/primary input
 - RedHawk does logic propagation to derive the activity information for internal logic
 - Mapping RTL VCD to gate netlist name
 - RedHawk can use auto-name mapping and generate the mapping on its own
 - Mapping file can be generated from formal verification tools
 - User control available for logic propagation through constraint file
-
- RTL VCD is useful during early stages of the design cycle
 - Gate level simulation happens later in the design cycle
 - RTL VCD file size will be much smaller compared to Gate level VCD
 - RTL VCD is always non true time
 - RedHawk will use switching scenario from VCD
 - RedHawk will use switching time from STA file



VCD Based Power Calculation



RedHawk VCD Based Dynamic Voltage Drop Analysis



Note : Fast Signal Database (FSDB) can also be used in place of VCD

VCD_FILE Specification in GSR

- **VCD_FILE Specification**

```
VCD_FILE
{
  <top_level name> <absolute or relative path to VCD or FSDB file>
  FILE_TYPE <VCD | FSDB | RTL_VCD | RTL_FSDB>
  FRONT_PATH <redundant path string that does not match the DEF path>
  SUBSTITUTE_PATH <the substitute path string from above>
  START_TIME <analysis start_time in ps; optional; default=0>
  TRUE_TIME [0|1]
  MAPPING <mapping_file_name>
}

DYNAMIC_SIMULATION_TIME <frame_width_sec>
```

- **Power Calculation will be done between:**
- **START_TIME and (START_TIME + DYNAMIC_SIMULATION_TIME)**

Hierarchical VCD_FILE Specification in GSR

- **BLOCK_VCD_FILE Specification**

`BLOCK_VCD_FILE`

```
{  
  VCD_FILE {  
    <hier_name/inst_name> <absolute or relative path to VCD or FSDB file>  
    FILE_TYPE <VCD | FSDB | RTL_VCD | RTL_FSDB>  
    FRONT_PATH <redundant path string that does not match the DEF path>  
    SUBSTITUTE_PATH <the substitute path string from above>  
    START_TIME <analysis start_time in ps; optional; default=0>  
    TRUE_TIME [0|1]  
    MAPPING <mapping_file_name>  
  }  
  VCD_FILE {  
    <hier_name2/inst_name2> <absolute or relative path to VCD/FSDB file>  
    FILE_TYPE <VCD | FSDB | RTL_VCD | RTL_FSDB>  
    .....  
  }  
}
```

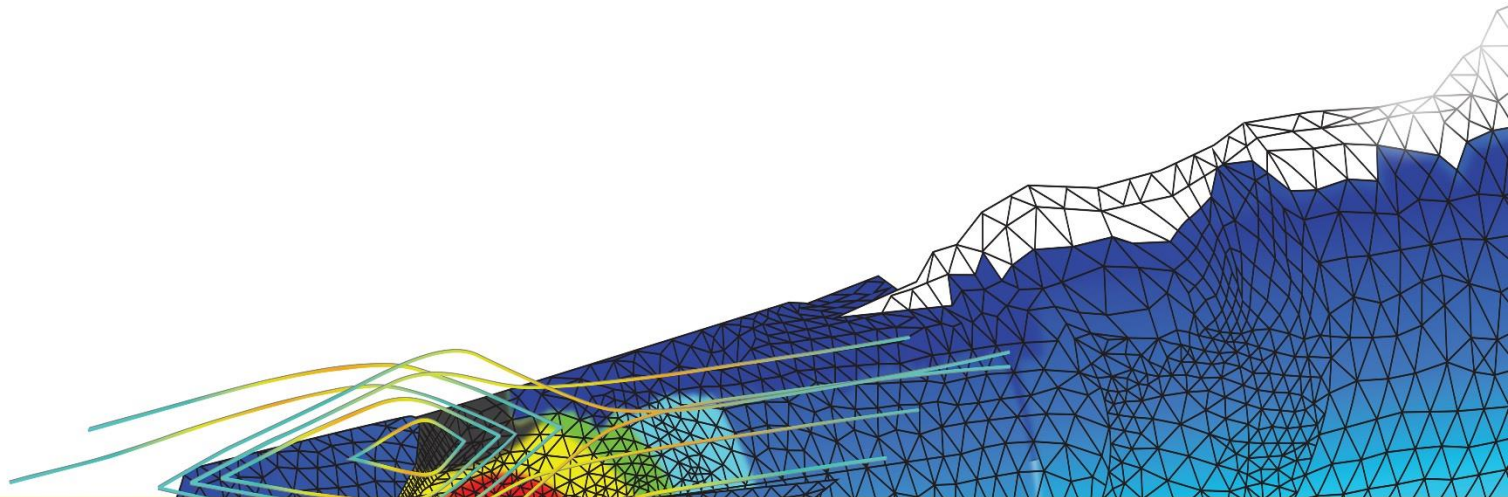
Power Calculation Options

POWER_ANALYSIS_MODE [averaged | averaged_fast]

- This allows user to set the accuracy level of VCD based power calculation. Setting the above keyword to “averaged” improves the accuracy of power calculation by taking into account state dependent leakage and internal power values from the .lib.
- **Default: averaged_fast.**

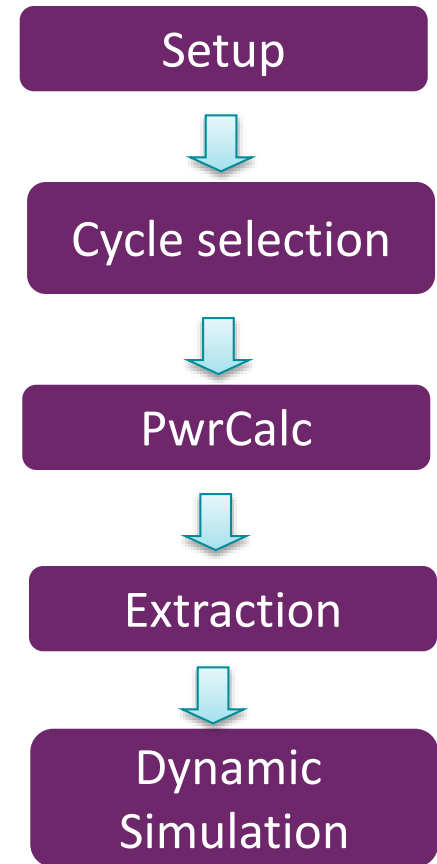


Cycle Select in VCD Based Dynamic Analysis

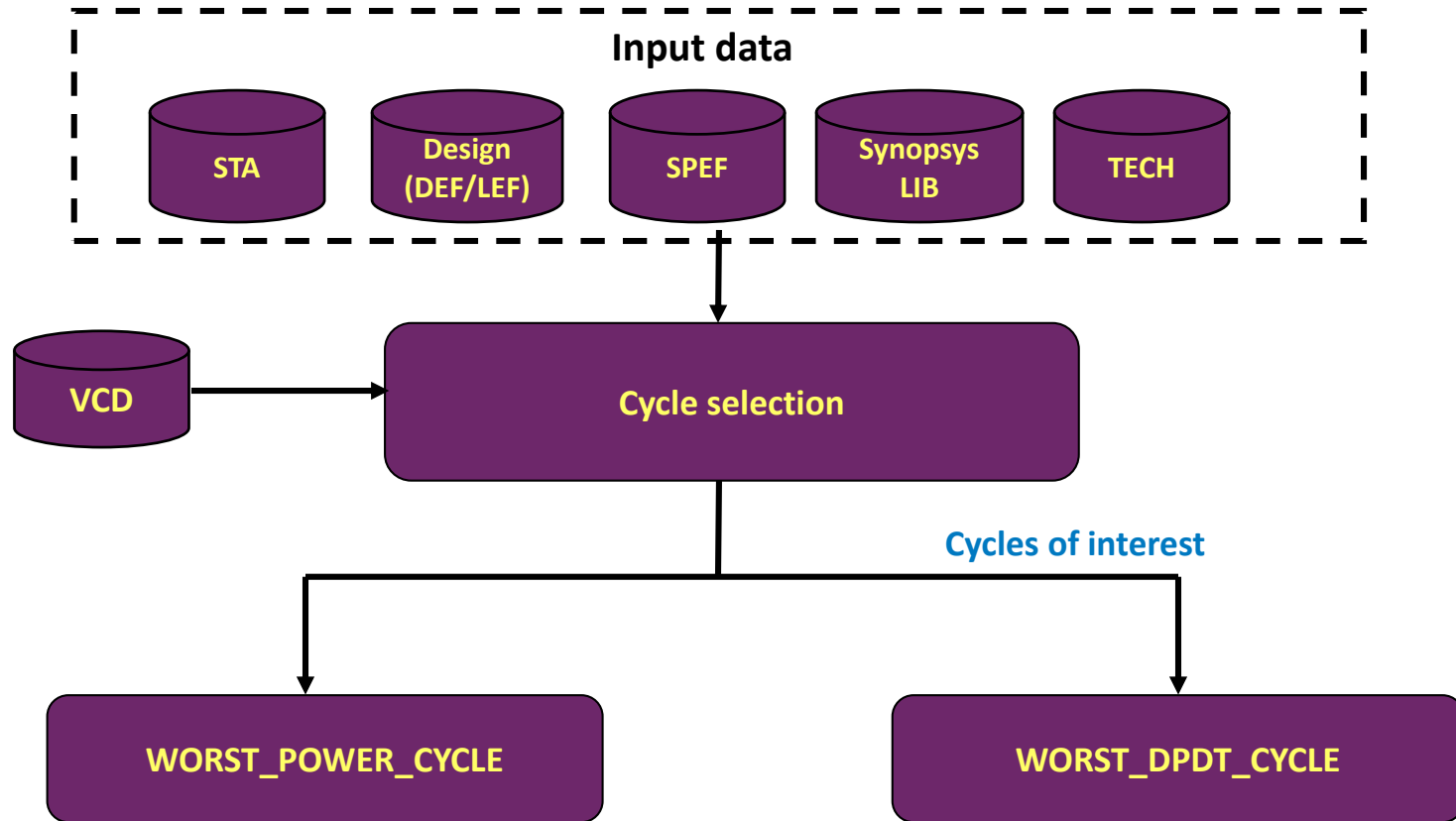


Cycle Select In VCD Based Dynamic Analysis

- **Challenges in VCD dynamic simulation**
 - Dynamic analysis is true transient simulation, not average analysis like static
 - VCDs can contain thousands of cycle. We cannot perform dynamic analysis for the complete VCD;
 - Need to intelligently filter out critical cycles in the VCD
- **Using Cycle selection to identify the critical cycles in the VCD**
- **Two methods to filter the critical cycles**
 - WORST_POWER_CYCLE
 - WORST_DPDT_CYCLE
- **Once the critical cycle is selected, RedHawk does dynamic simulation for this cycle**
 - Pre-simulation cycles (from VCD) will be attached to the selected cycle
- **Both True-time and Non-true time VCDs can be used in cycle selection**
- **Both RTL and Gate level VCDs can be used for cycle selection**
- **The keyword SELECT_RANGE activates cycle selection.**
 - The VCD time between the values provided in SELECT_RANGE will be considered for cycle selection.



Cycle Select In VCD Based Analysis Flow



```
VCD_FILE {  
  SELECT_RANGE <start_time> <end_time>  
  SELECT_TYPE WORST_POWER_CYCLE  
}
```

```
VCD_FILE {  
  SELECT_RANGE <start_time> <end_time>  
  SELECT_TYPE WORST_DPDT_CYCLE  
}
```

WORST_POWER_CYCLE

GSR and Command file examples

GSR:

```
VCD_FILE {  
  GENERIC GENERIC.vcd  
  FILE_TYPE VCD  
  FRONT_PATH "GENERIC/"  
  SUBSTITUTE_PATH ""  
  SELECT_RANGE <start_time> <end_time>  
  SELECT_TYPE WORST_POWER_CYCLE  
  TRUE_TIME 1  
}
```

TCL:

```
import gsr GENERIC.gsr  
setup design  
setup analysis_mode dynamic  
perform pwrcalc  
perform extraction -power -ground -c  
perform analysis -vcd
```

- **DYNAMIC_SIMULATION_TIME** decides the window width of each cycle
- User can set "**SELECT_RANGE -1 -1**" to allow RedHawk to do a cycle selection for the entire VCD range

WORST_POWER_CYCLE (Cont'd)

- RedHawk does cycle by cycle power calculation

adsRpt/worst_power_cycle.rpt

```
# VCD_FILE ../../../../design_data/vcd/GENERIC.vcd
# BLOCK_NAME GENERIC
# SELECT_RANGE 0 5000
# SELECT_TYPE WORST_POWER_CYCLE
# FRAME_SIZE 1280
# WINDOW_GRIDS 2

# cycle power rank (sorted by average_power)
#      cycle_time(ps)      average_power(W)
1      0 to      2560      1.45887
2     1280 to     3840      1.33492
3     2560 to     5120      0.64573
```

- Selects cycle having highest power for dynamic simulation

WORST_DPDT_CYCLE

GSR and Command file examples

GSR:

```
VCD_FILE {  
  GENERIC GENERIC.vcd  
  FILE_TYPE VCD  
  FRONT_PATH "GENERIC/"  
  SUBSTITUTE_PATH ""  
  SELECT_RANGE <start_time> <end_time>  
  SELECT_TYPE WORST_DPDT_CYCLE  
  TRUE_TIME 1  
}
```

TCL:

```
import gsr GENERIC.gsr  
setup design  
setup analysis_mode dynamic  
perform pwrcalc  
perform extraction -power -ground -c  
setup package  
perform analysis -vcd
```

- DYNAMIC_SIMULATION_TIME decides the window width of each cycle.
- User can set "SELECT_RANGE -1 -1" to allow RedHawk to do a cycle selection for the entire VCD range.

Accuracy In Cycle Selection Based Power Calculation

- **Accuracy can be selected using the below keyword:**
 - `POWER_CYCLE_SELECT_MODE [0 | 1 | 2]`
 - 0: default value, time usage: low, memory usage: low, accuracy: low
 - 1: time usage: low, memory usage: high, accuracy: middle
 - 2: time usage: high, memory usage: high, accuracy: high

Result Exploration and Debugging

```
*****
****   Start Cycle Selection
*****

INFO: Consider memory switching states from VCD file(s) in p
Running vcdscan ../../design_data/vcd/GENERIC.vcd -tt -w
5000p -f 1280p -o .apache/cfilter.0.pwr -ap NOPRINT -qs
Reading ../../design_data/vcd/GENERIC.vcd ...
Turn on module handle on begin/fork.
ERROR(EVP-164): VCD symbol 'inst_129422/net_4412' is defined more than once, ignored.
ERROR(EVP-164): VCD symbol 'inst_129422/net_3418' is defined more than once, ignored.
ERROR(EVP-164): VCD symbol 'inst_129422/net_3418' is defined more than once, ignored.
  0 memory with control defined
Traversing transitions ...
..5%WARNING(VCD-311): For instance inst_129539/adsU1, state c10 at 400ps, is too close to previ
ous state c01 (less than 0.4*cycles). This state will be ignored.
..11% ..16% ..22% ..27% ..32% ..38% ..43% ..49% ..54% ..59% ..64% ..70% ..75% ..80% ..85% ..90%
..96% ..100%
WALLTIME (Traversing transitions): 5 secs

Summary for VCD/FSDB file ../../design_data/vcd/GENERIC.vcd:
VCD/FSDB start time (1 ps) : 0
Sim start time (ps) : 0
Pre-sim start time (ps) : 0
Sim end time (ps) : 5000
CPU TIME (vcdscan): 0 hrs 0 mins 9 secs
MEMORY USAGE (vcdscan): 383 MBytes

***** Cycle selection summary *****
Selection resolution (grid size) is 1280ps, total cycle 3
Selection from 0ps to 5000ps, total grids 4
Average power of the entire selection range: 1.14651 W
```

adsRpt/redhawk.log

1422,2

63%

Debug Tips in VCD Analysis

- **Check whether the VCD is imported correctly in RedHawk**
 - adsRpt/vcd_missing_signal_nets
 - adsRpt/vcd_missing_clock_nets
- **If close to 100% of nets are reported as missing in the VCD check the “FRONT_PATH” keyword in GSR**
 - Make sure that there is a trailing slash in the string specification
 - FRONT_PATH "test_bench1/GENERIC/"
 - Open the VCD in text editor and verify the scope statements
 - Scope statements define the hierarchy in VCD
 - If the input is FSDB, you can convert it into VCD using 3rd party conversion utilities and check
 - Load the VCD/FSDB in a 3rd party waveform viewer to browse the hierarchy
- **Use switching histograms to verify the VCD switching**
 - For scan shift VCDs, make sure that ~100% flops and clocks are switching
 - For MBIST VCDs, make sure that ~100% memories are switching
- **For true-time VCD analysis, make sure that VCD is timing back annotated**

Lab Exercise : VCD Analysis

- **Steps to be followed**

- cd GENERIC_tutorial/vcd_run
- cd worst_power
- Review GSR and command file
- redHawk -f run_dynamic.tcl

User control file (GSR):

```
VCD_FILE {  
  GENERIC ../../design_data/vcd  
            /GENERIC.vcd  
  FILE_TYPE VCD  
  FRONT_PATH "test_bench1/GENERIC/"  
  TRUE_TIME 1  
  SELECT_RANGE -1 -1  
  SELECT_TYPE WORST_POWER_CYCLE  
}
```

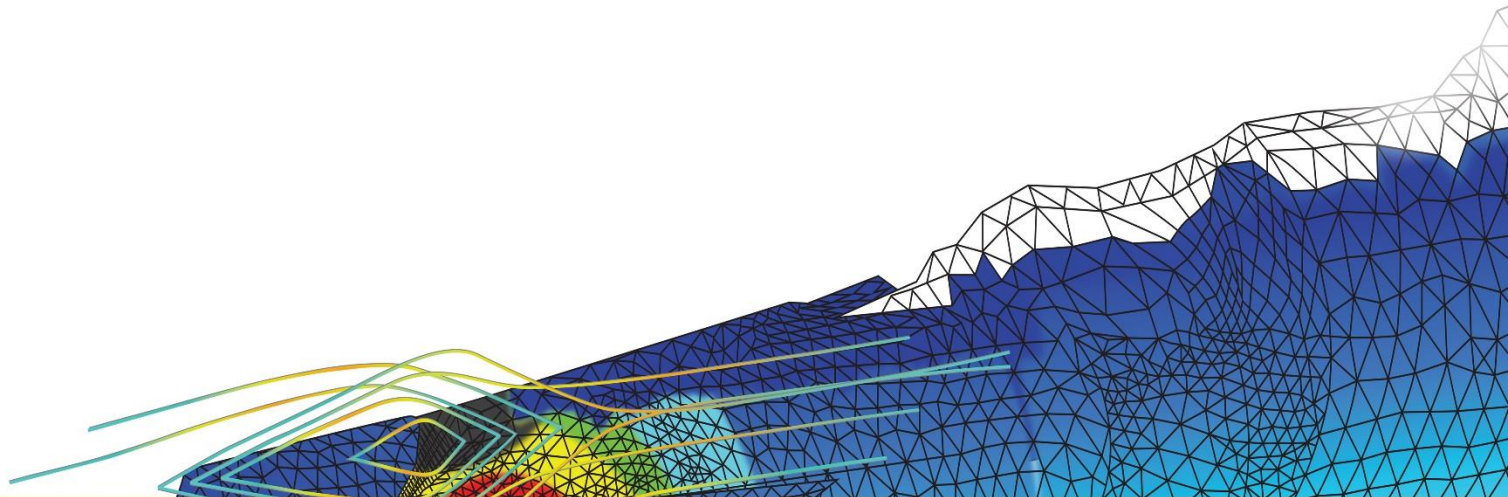
```
# Import data  
import gsr GENERIC.gsr  
setup design  
setup analysis_mode dynamic  
  
# Calculate power  
perform pwrcalc  
  
# Power grid extraction  
perform extraction -power -ground -c  
  
# Lumped resistance (in Ohms)  
# for package, wirebond and pads  
# Default values are all 0, i.e no off-chip impact  
setup package -power -r 0.005 -l 2.5 -c 5  
setup package -ground -r 0.005 -l 2.5 -c 5  
setup wirebond -power -r 0.01 -l 2.2 -c 1.42  
setup wirebond -ground -r 0.05 -l 1.7 -c 0.2  
setup pad -power -r 0.001  
setup pad -ground -r 0.001  
  
# Dynamic IR analysis  
perform analysis -vcd  
  
explore design
```

Result Analysis

- **Result analysis procedure is same as normal dynamic analysis**
 - Refer to Lecture : “Moving to RedHawk Dynamic”



Advanced Usage Options



RTL-VCD Auto Name Mapping

- To perform RTL-to-gate mapping of net names . Default ON in RedHawk
- GSR and Command file examples

GSR:

```
VCD_FILE {  
  GENERIC GENERIC.vcd  
  FILE_TYPE RTL_VCD  
  FRONT_PATH "GENERIC/"  
  SUBSTITUTE_PATH ""  
  START_TIME <start_time_in_ps>  
  TRUE_TIME 0  
}
```

TCL:

```
import gsr GENERIC.gsr  
setup design  
setup analysis_mode dynamic  
perform pwrcalc  
perform extraction -power -ground -c  
setup package  
perform analysis -vcd
```

- To turn off RTL-VCD Auto Name Mapping enable 'RTL_NAME_MAPPING 0'
in GSR

RTL-VCD Propagation Modes

- Event Propagation Based Power Calculation (default):

GSR file

```
VCD_FILE {  
  GENERIC GENERIC.vcd  
  FILE_TYPE RTL_VCD  
  START_TIME <start_time_in_ps>  
  .....  
}
```

No special settings needed – Event Propagation is the default mode when RTL VCD is specified.

- State Propagation Based Power Calculation :

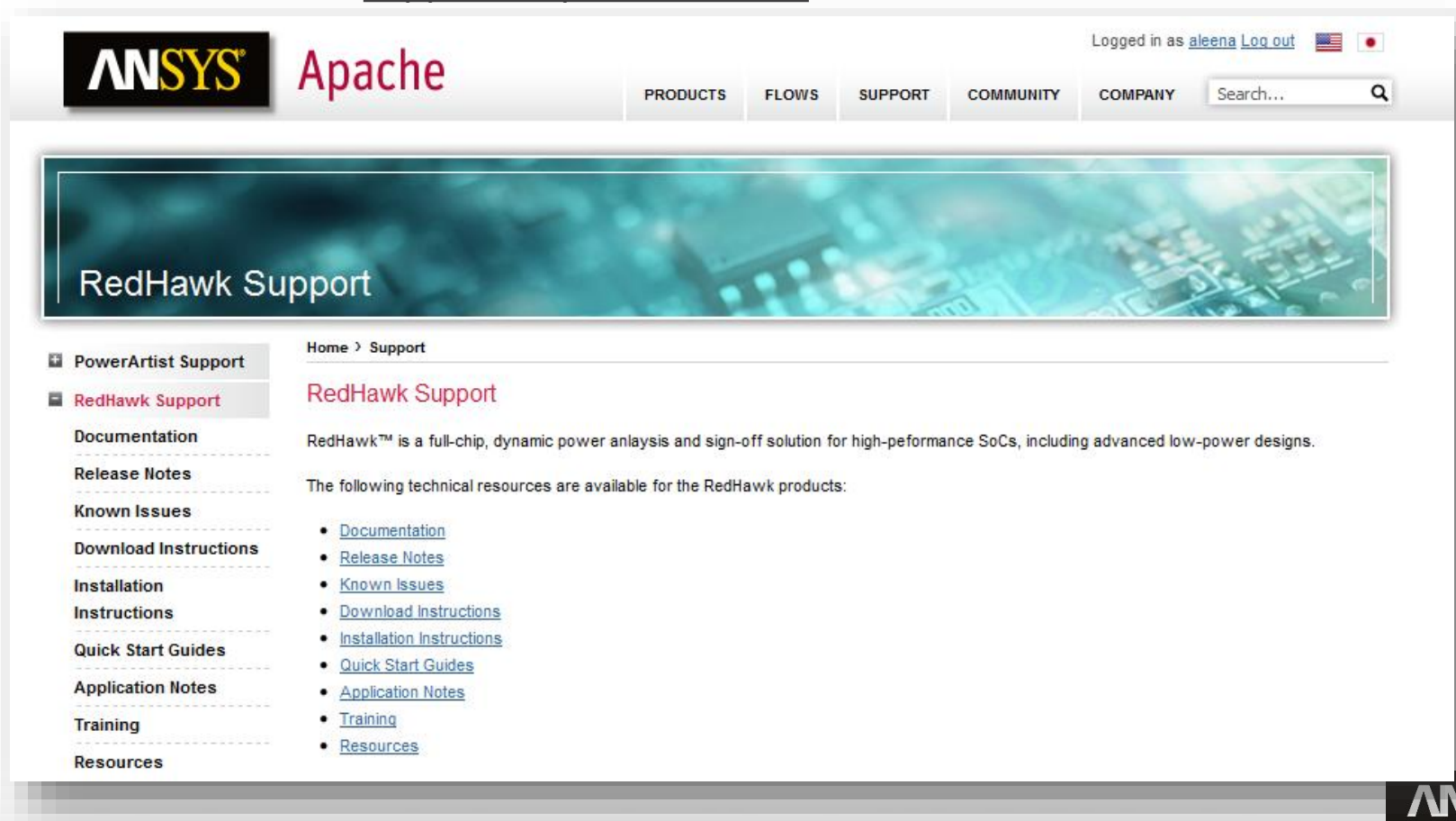
GSR file

```
VCD_FILE {  
  GENERIC GENERIC.vcd  
  FILE_TYPE RTL_VCD  
  START_TIME <start_time_in_ps>  
  .  
  .  
  VCD_DRIVEN 1  
}
```

- Keyword '**VCD_DRIVEN 1**' enables State Propagation.
- Note: END_TIME is honored here. VCD is analyzed from START_TIME to END_TIME.

How to Get Help!

- **Apache Online Customer Support Center**
 - <http://support.apache-da.com>
 - Email: support@apache-da.com



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Thank You!